

# VINAY K R

Software Engineer | Full-Stack, Backend, Applied AI & Reliability

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## SUMMARY

Software Engineer with nearly three years of professional experience delivering full-stack applications and APIs in fintech and regulated gaming. Professional strengths include Angular, TypeScript/Node.js, C#/.NET, REST APIs, Redis, RBAC, automated testing, and production debugging. Public projects add hands-on Go and Python/FastAPI development, LangGraph and RAG, plus Linux, Docker, Kubernetes labs, Terraform, Prometheus, and Grafana.

## TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Go, Python, C#, SQL, Bash
- **Application Engineering:** Angular, React, RxJS, Node.js, Express.js, ASP.NET Core, FastAPI, REST APIs, WebSockets, microservices
- **Data & Security:** PostgreSQL, Redis, SQL Server, Auth0, RBAC, multi-tenancy, JWT authentication
- **Applied AI:** LangGraph, agentic workflows, RAG, hybrid retrieval, BM25, vector search, embeddings, Pydantic, OpenTelemetry
- **Platform & Quality:** Linux, Docker, Kubernetes, Terraform, Ansible, AWS fundamentals, GitHub Actions, Prometheus, Grafana, Cypress, pytest

## EXPERIENCE

**Software Engineer — Full Stack** | Liminal Custody (First Answer India Services Pvt Ltd) Nov 2025 – Mar 2026

- Built Angular workflows for Transaction Risk, Travel Rule, and Transfer Policy controls using RxJS state, cascading inputs, reusable components, and unsaved-change guards.
- Developed Node.js/Express REST APIs for firewall policy management with atomic bulk writes and transaction rollbacks, protecting consistency during concurrent rule updates.
- Contributed to Node.js-to-Go modernization of Bitcoin input resolution, replacing repeated upstream RPC patterns with an indexed UTXO lookup path.
- Implemented multi-tenant RBAC and quorum flows across Auth0, Angular guards/interceptors, and backend middleware, including step-up authentication and signature validation for sensitive endpoints.
- Built Redis caching for cross-chain quote data across Ethereum, Polygon, BSC, and Base, validating ownership and expiry to reduce avoidable external API calls.

**Senior Associate Software Engineer** | Light & Wonder (LNW India Solutions Pvt Ltd) Aug 2023 – Jul 2025

- Developed modular Angular components for player-management workflows and optimized relational queries and REST API payloads for responsive data-heavy screens.
- Built React device widgets supporting attendant calls, beverage orders, and reservations across three user interaction flows.
- Implemented C#/.NET Core Web API features for bank accounts, voucher redemption, and check-cashing workflows in a regulated gaming environment.
- Diagnosed and fixed an RxJS/WebSocket subscription-lifecycle leak in continuously operating gaming-device software, improving runtime stability.
- Built a React and LLM-assisted debugging tool that centralized frontend crash reports and backend telemetry; maintained Cypress end-to-end tests to strengthen release confidence.

**Full-Stack Intern** | Light & Wonder (LNW India Solutions Pvt Ltd) Mar 2023 – Jul 2023

- Completed a 16-week C#/.NET Core and Angular internship and delivered a game-recommendation application with user feedback and ratings.

## PROJECTS

**Medha Platform** | Go, Linux, Docker, PostgreSQL, Redis, Nginx

- Built Go REST/WebSocket APIs for four client surfaces with Chi, PostgreSQL/PostGIS, Redis Pub/Sub, and SeaweedFS.
- Used Handler-Service-Repository layers, pgx transactions, RFC 7807 errors, OTP/RS256 JWTs, request IDs, and graceful shutdown.
- Operate a two-node Debian/Docker platform with systemd, Nginx, health checks, structured logs, verified backups, and rollback.

**FastAPI GenAI Agent Patterns** | Python, FastAPI, LangGraph, Redis 8, OpenTelemetry

- Build a provider-neutral FastAPI service with typed LangGraph specialist routing, bounded iterations, human approval, and SSE.
- Implemented two-stage Redis 8 caching with SHA-256 exact keys, vector search, local embeddings, and cache statistics.
- Exported OpenTelemetry traces for API and agent paths to Jaeger and validated behavior in GitHub Actions CI.

**Enterprise Agentic RAG Platform** | Python, FastAPI, LangGraph, Hybrid Retrieval, RRF

- Built document ingestion and LangGraph retrieval/tool routing with relevance grading, query rewriting, and cited answers.
- Combined vector and BM25 search using reciprocal rank fusion, context reranking, and compression.
- Added JSON-RPC database/API tools and automated faithfulness, relevance, and grounding evaluation.

**Reliability Engineering Labs** | Kubernetes, Prometheus, Grafana, Terraform, Ansible

- Instrumented Go availability/latency SLIs, a 99.5% SLO, Prometheus burn alerts, Grafana dashboards, and black-box probes.
- Automated seven Kubernetes failure scenarios and validated safe rollout, rollback, recovery evidence, and postmortems.
- Built idempotent Ansible Linux baselines and a cost-bounded AWS Terraform reference with ECS, ALB, alarms, and budgets.

## EDUCATION & RESEARCH

**B.E. Computer Science & Engineering**, Siddaganga Institute of Technology | 2019–2023 | CGPA 8.65/10

Co-author, “Data Visualisation of Time Tradable Assets Using Machine Learning,” IEEE, 2023